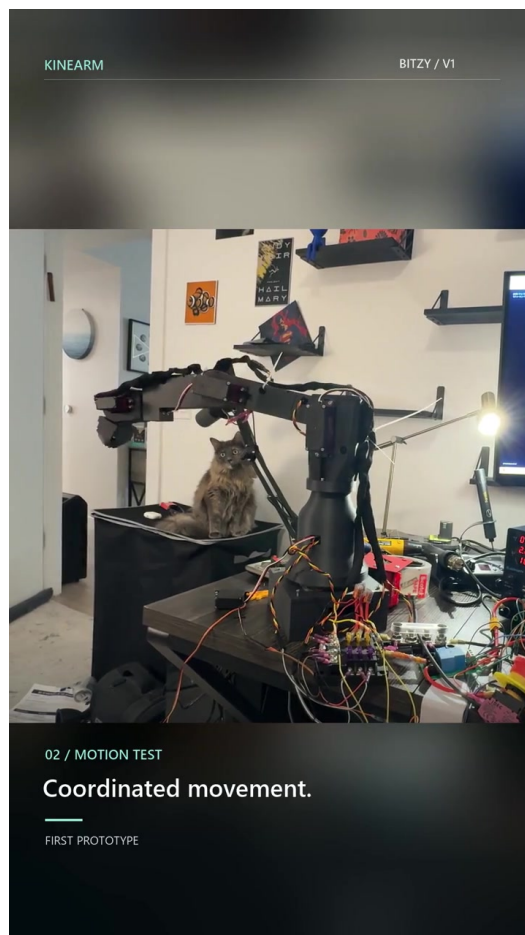


Bitzy V1

Modular Robotic Arm

Harrison B. Goldberg / June 2026 - Present

Built prototype; control refinement and physical qualification ongoing.



Frame from the physical prototype demo. Some assembly segments in the video are sped up.

A built five-axis robotic arm combining printed mechanics, custom embedded control, inverse kinematics and interchangeable tooling. The project progresses from an earlier six-axis study to active hardware and recorded motion workflows.

AI-assisted workflows supported the creation of this project.

System design

The five-axis arm uses six arm servos, including a synchronized shoulder pair. The current tool interface is retained across the gripper and screwdriver revisions.

System	Reference
Arm architecture	Five controlled arm axes; six arm servos because the shoulder uses a synchronized pair
Current servo record	DS5160SSG shoulder pair and elbow; MS24 replacements reported at base yaw and both wrist axes
Motion/control	ESP32-S3, PCA9685, Python desktop bridge and browser controller; MATLAB/Python kinematics
Nominal reach reference	Approximately 475 mm; design reference, not an independently measured workspace
Payload objective	500 g object plus tool mass; target only, no qualified load envelope
Shared tool interface	Original star and four-screw mounting interface retained
Gripper Rev B	108 named parameters, 84 CAD components; 2-26 mm nominal soft-jaw opening, 12 mm travel per jaw
Gripper packaging	74 x 73 mm nominal case; PETG-CF case, replaceable TPU sleeves; 25 product shapes and six fit coupons
Screwdriver R2	136 named parameters, 67 CAD components; 4:1 external reduction, 3 mm guided axial compliance
Screwdriver packaging	80 x 70 mm nominal case; 12 product prints and 11 fit ladders; 287.7 g estimated installed mass, 340 g planning allowance
Tool attachment	Manual exchange through the common mounting interface; automatic docking is future work

Approximate reach and payload objectives are nominal design references. The current hardware has no published quantitative accuracy or payload qualification.

Evidence and limitations

Review	Recorded evidence	Remaining qualification
Physical arm	Built hardware and movement shown in the demo	Demonstration does not quantify payload, accuracy, repeatability or duty.
Teach/replay	Named presets, automatic travel, synchronized cubic joint interpolation and quintic start/stop easing recorded	Commanded/model pose is not a measured physical pose; validate loaded replay on hardware.
Shoulder limit	User-reported back stop near -29.87 degrees informed a -28 degree command guard	A local limit correction does not establish whole-arm collision freedom.
Software regression	28 Python checks plus JavaScript control/replay suites passed in the September 9 record	Recorded software checks do not establish physical positioning or hardware reliability.
Gripper Rev B	Nine native travel positions and 181 exported gear-section positions checked; 31 meshes sliced	Actual grip force, printed fits, friction, wear and arm loads remain unqualified.
Screwdriver R2	11 native poses, six rotation transforms and 1,441 gear-section checks; 23 meshes sliced	Actual bit retention, runout, torque, current, duty and physical fits remain unqualified.
Reliability	Intermittent joint/serial behavior remains an open investigation	Spontaneous recovery and replay changes do not establish the original electrical fault cause.

Video demonstrates assembly and movement. Software regressions validate command behavior. CAD/mesh checks validate digital artifacts. None substitutes for physical metrology and loaded endurance trials.

Budget and next stage

The arm is built; total V1 build cost is not reconciled. The historical power/wiring list mixes owned hardware, ordered packs, deferred equipment and optional supplies.

Selected historical price reference	USD
ESP32-S3 board	\$18.99
Latching emergency-stop button	\$14.99
12 AWG wire pair, 25 ft	\$34.48
Dedicated 6 V supply (deferred)	\$199.99

Recorded August 31 / September 1, 2026. Partial item quotes before tax and shipping; not current prices, full-build spend or a complete BOM.

Next engineering work

Document the exact as-built revision. Qualify the gripper and screwdriver, establish actual tool-centre locations and tool mass, measure repeated approaches and loads, and correlate intermittent joint response with electrical and serial behavior.

Bitzy V1 continues independently. The proposed motor-driven successor is planning work and does not change this prototype record.

Selected public release

Assembly review (2 pages), tooling and service (2 pages), and control/validation (3 pages) accompany this portfolio. Physical video, actual CAD views and historical kinematic sketches are identified by revision. Editable CAD, manufacturing and control sources remain private.

AI-assisted workflows supported the creation of this project.